# DebiCheck Instalment Maintenance Specifications Document

# DRAFT

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|---|---|---|---|
| V1.0. | 2019/09/11 | Brandon Velloo | Initial Draft |
| V1.1. | 2019/10/01 | Brandon Velloo | Update to input field descriptions |
| V1.2. | 2021/02/22 | Brandon Velloo | Addition of “applytoAll” field |
| V1.3. | 2022/05/19 | Nolwazi Mokgobinyane | “applytoAll” field changes |
| V1.4. | 2023/03/24 | Danielle Haylett | Addition of Re-activate instalment method |
| V1.5. | 2024/05/16 | Danielle Haylett | Addition of Recall instalment |

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# 1. Disclaimer

The DebiCheck payment stream is still under development, thus this document is subject to change and it is not the final draft.

# 2. Introduction

This document describes the DebiCheck full function webservices for instalment maintenance:

- Add an instalment
- Maintain an instalment
- Reschedule an instalment
- Cancel an instalment
- Re-activate an instalment
- Recall an instalment

## 2.1. System Overview

The merchant will be sending a request type to NuPay via webservices. NuPay needs to accept the request and authenticate the merchant using the authentication string. Once authentication has taken place the data sent through will be processed. All requests are handled by the below mentioned methods of the webservice.

## 2.2. Security

Security measures include the use of Secure Sockets Layer (SSL) for the connection. It is required to use https instead of the normal http when connecting to the server.

## 2.3. Best Practice for Webservice Consumption

The webservices adhere to the Restful standard, which dictates that the input and output is Json data. RESTful web services are built to work best on the web.

Representational State Transfer (REST) is an architectural style that specifies constraints, such as the uniform interface, that if applied to a web service induce desirable properties, such as performance, scalability, and modifiability, that enable services to work best on the web.

The suggested method of using this web service is to supply a JSON call with all the input parameters.

# 3. Connections

## 3.1. Add Instalment

URL: https://btm.nupay.co.za/wsDebiCheck/add_instalment

## 3.2. Instalment Maintenance

URL: https://btm.nupay.co.za/wsDebiCheck/instalment_maintenance

## 3.3. Instalment Reschedule

URL: https://btm.nupay.co.za/wsDebiCheck/instalment_reschedule

## 3.4. Instalment Cancellation

URL: https://btm.nupay.co.za/wsDebiCheck/instalment_cancellation

## 3.5. Instalment Re-Activate

URL: https://btm.nupay.co.za/wsDebiCheck/reactivate_instalment

## 3.6. Instalment Recall

URL: https://btm.nupay.co.za/wsDebiCheck/instalment_recall

# 1. System Inputs

## 1.1. Add Instalment

This method adds a new instalment to the existing mandate. All the inputs below are required. The submit date must be in line with the original mandate. The mandate must be active.

The following inputs apply in JSon format:

```
{
        "auth":"",
        "mandateID":"",
        "cardAcceptor":"",
        "instalment":"",
        "amount":"",
        "trackingIndicator":"",
        "submitDate":""
}
```

## 1.2. Instalment Maintenance

This method allows maintenance to be done on an instalment. Only the amount, tracking indicator and submit date can be maintained on an instalment. Only active future instalments can be maintained. The following inputs apply in JSon format:

```
{
        "auth":"",
        "mandateID":"",
        "cardAcceptor":"",
        "instalment":"",
        "amount":"",
        "trackingIndicator":"",
        "submitDate":"",
      "applyToAll":
}
```

### 1.3. Instalment Reschedule

This method allows an instalment to be rescheduled. Only a failed instalment can be rescheduled. All inputs are required. The date and the tracking of the failed instalment can also be updated.

The following inputs apply in JSon format:

```
{
        "auth":"",
        "mandateID":"",
        "cardAcceptor":"",
        "instalment":"",
        "amount":"",
        "trackingIndicator":"",
        "submitDate":""
}
```

### 1.4. Instalment Cancellation

This method allows an instalment to be cancelled. Only active future instalments can be cancelled.

The following inputs apply in JSon format:

```
{
        "auth":"",
        "mandateID":"",
        "cardAcceptor":"",
        "instalment":"",
        "amount":"",
        "trackingIndicator":"",
        "submitDate":""
}
```

## 1.5. Instalment Re-activate

This method allows a cancelled instalment to be re-activated. Only future instalments can be reactivated for their original submit date.

The following inputs apply in JSon format:

```
{
        "auth":"",
        "mandateID":"",
        "accessID":"",
        "instalment":"",
        "startDate":""
}
```

## 1.6. Instalment Recall

This method allows an instalment to be recalled. Only instalments that are in an 'In Tracking' state can be recalled.

The following inputs apply in JSon format:

```
{
        "auth":"",
        "cardAcceptor":"",
        "mandateID":"",
        "instalment":""
}
```

## 1.7. Input Field Descriptions

| Data Element | Description | Data Type and Length |
|---|---|---|
| Auth | Username and password in format “username:password” and Base64Encoded | String |
| Mandate ID | Mandate ID. For example, 133068 | String |
| Card acceptor / Access ID | Merchant number. Must be zero padded and pre-registered at NuPay.<br>For example, 000005500000010 | String (15) |
| Instalment | The instalment number in the mandate. | String |
| Amount | The amount of the instalment. This value is in cents. For example, 100 for an instalment amount of R1.00. | String |
| Tracking Indicator | The number of tracking days. Must be 01-10 corresponding to 1 – 10 days respectively. | String (2) |
| Submit date /Start date | Instalment submission date.<br>Format must be YYYYMMDD. | String (8) |
| Apply to All | **Only applies to tracking indicator and submit date.**<br><br>The following inputs are allowed:<br>• True<br>• False<br>If true, will apply the change to all instalments. If false, will apply to selected instalment only.<br><br>The field may be excluded completely from the request, in which case it will default to false. If the field is included in the request, either true or false must be entered. | Boolean |

# 2. System Outputs

## 2.1. Add instalment

The following outputs apply in JSon format:

```
{
    "cardAcceptor": "",
    "mandateID": "",
    "instalment": "",
    "instalmentAmount": "",
    "trackingIndicator": "",
    "cycleDate": "",
    "submitDate": "",
    "disputable": "",
    "instalmentStatus": "",
    "responseCode": "",
    "responseDescription": ""
}
```

## 2.2. Instalment Maintenance

The following outputs apply in JSon format:

```
{
    "cardAcceptor": "",
    "mandateID": "",
    "instalment": "",
    "instalmentAmount": "",
    "trackingIndicator": "",
    "cycleDate": "",
    "submitDate": "",
    "disputable": "",
    "instalmentStatus": "",
    "responseCode": "",
    "responseDescription": ""
}
```

## 2.3. Reschedule Instalment

The following outputs apply in JSon format:

```
{
   "cardAcceptor": "",
   "mandateID": "",
   "instalment": "",
   "instalmentAmount": "",
   "trackingIndicator": "",
   "cycleDate": "",
   "submitDate": "",
   "disputable": "",
   "instalmentStatus": "",
   "responseCode": "",
   "responseDescription": ""
}
```

## 2.4. Instalment Cancellation

The following outputs apply in JSon format:

```
{
   "cardAcceptor": "",
   "mandateID": "",
   "responseCode": "",
   "responseDescription": ""
}
```

## 2.5. Instalment Re-activate

The following outputs apply in JSon format:

```
{
   "trackingIndicator": "",
   "disputable": "",
   "instalmentStatus": "",
   "instalment": "",
   "responseDescription": "",
   "mandateID": "",
   "cardAcceptor": "",
   "instalmentAmount": "",
   "submitDate": "",
   "cycleDate": "",
   "responseCode": ""
}
```

## 2.6. Instalment Recall

The following outputs apply in JSon format:

```
{
   "responseDescription": "",
   "mandateID": "",
   "cardAcceptor": "",
   "responseCode": ""
}
```

## 2.7. Output Field Descriptions

| Data Element | Description | Data Type and Length |
|---|---|---|
| Card acceptor | Merchant number. Zero padded. | String (15) |
| Mandate ID | Mandate ID | String |
| Instalment | Instalment number in the mandate. | String |
| Instalment amount | Instalment amount. Returned in cents. | String |
| Tracking indicator | 01-10 corresponding to 1 – 10 days respectively | String (2) |
| Cycle date | Instalment cycle date | String |
| Submit date | Instalment submit date | String |
| Disputable | 1 = yes<br>0 = no | String (1) |
| Instalment status | X = success<br>H = failed<br>P = future<br>I = in tracking<br>C = cancelled | String (1) |
| Response code | Refer to appendix 6.1. | String |
| Response description | Refer to appendix 6.1. | String |

# 3. Appendix

## 3.1. Error Codes

| Code | Description |
|---|---|
| 500000 | Success |
| 500002 – 13 | Authentication Failed: User Currently Blocked. |
| 500002 – 14 | Authentication Failed: Too many failed attempts; User Blocked. |
| 500002 – 15 | Authentication Failed: No User found or Incorrect Details. |
| 500002 – 20 | Authentication Failed: Please insert \":\" between username and password before encoding.zzzz |
| 500002 – 21 | Authentication Failed. |
| 500002 – 23 | Auth cannot be null or empty. |
| 500002 | User Credentials not linked to merchant. Please use different credentials. |
| 500002 | Generated token is invalid. |
| 500002 | Parse exception in ‘method name’. |
| 500002 | No Data for Disputed Reports. |
| 500007 | Internal Error |
| 500039 | Mandate not found |
| 500040 | Change not applied |
| 500041 | Invalid mandate_id format when attempting maintenance |
| 500042 | Change not allowed on First Collection |
| 500043 | General error |
| 500048 | Incorrect status on instalment |
| 500049 | Incorrect status on mandate |
| 901111 | Instalment amount is greater than the Maximum Collection Amount |